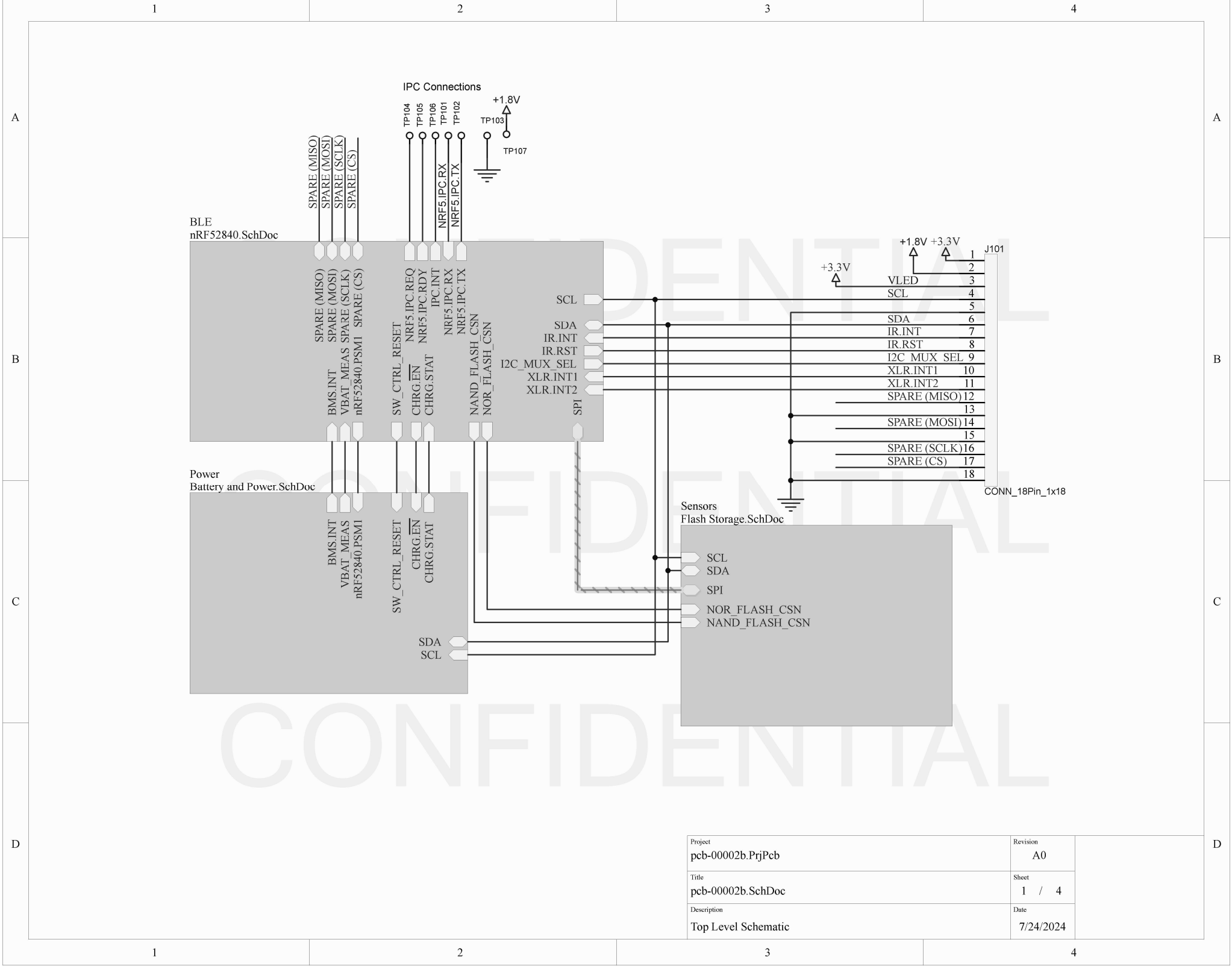
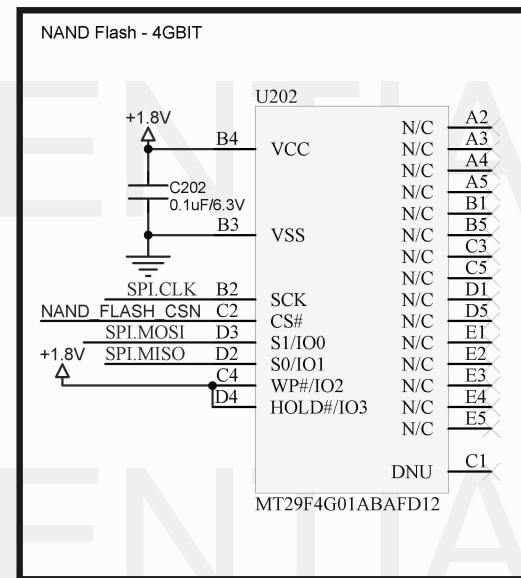
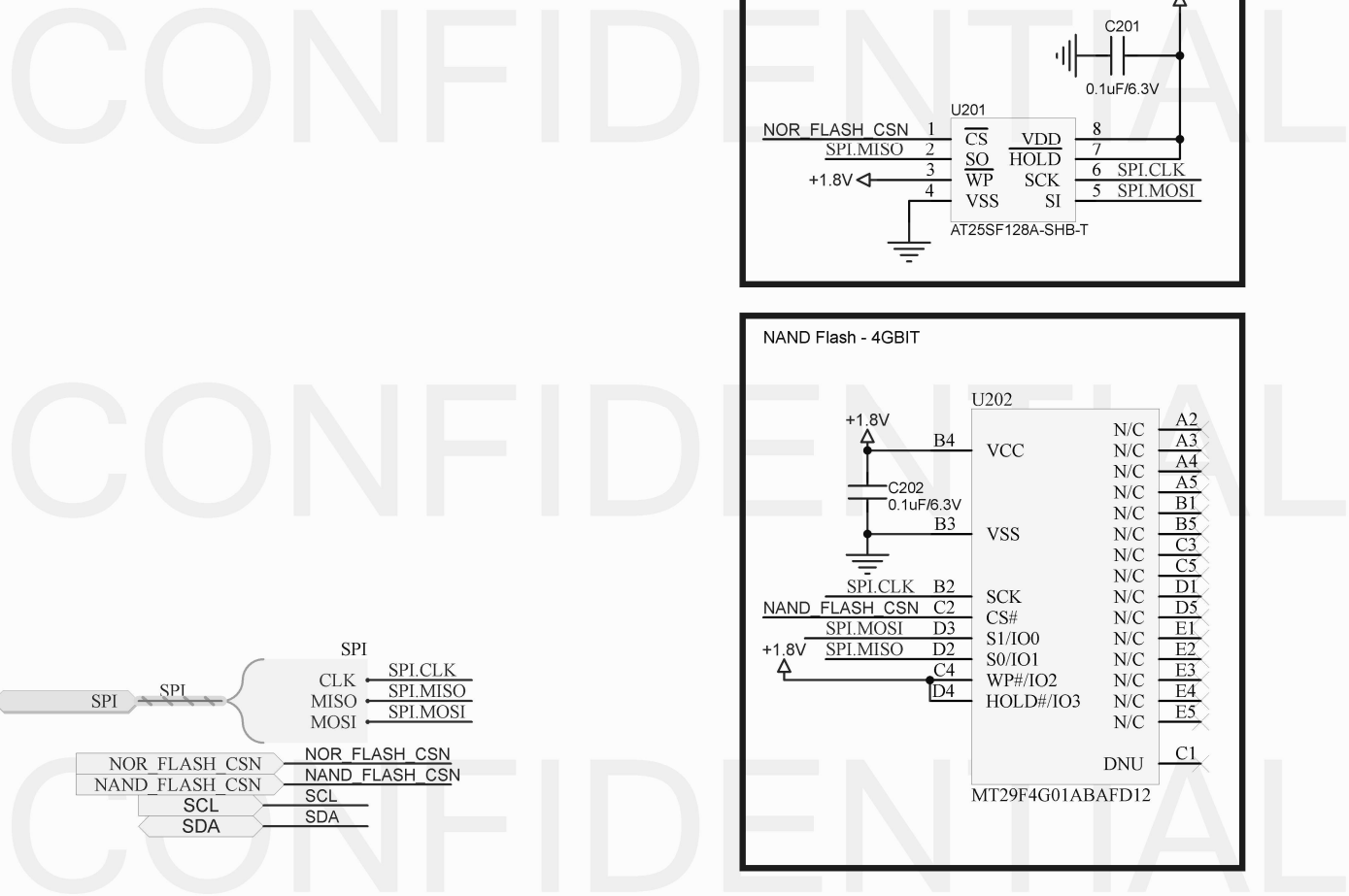
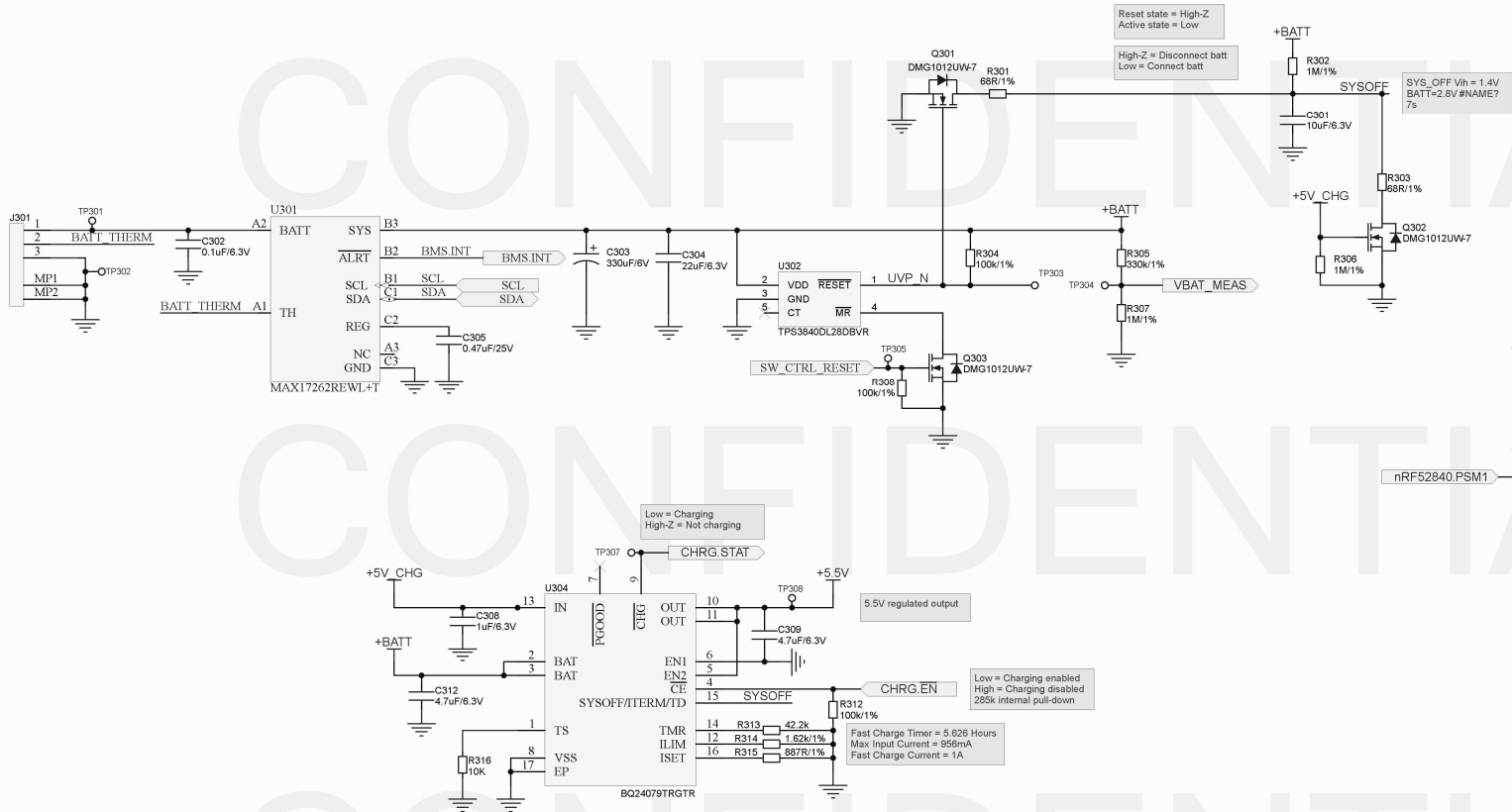






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- 12 Layout**
- 12.1 Layout Guidelines**
- The PCB layout is an important step to maintain the high performance of the TPS63802 device.
1. Place input and output capacitors as close as possible to the IC. Traces need to be kept short. Route wide and direct traces to the input and output capacitor results in low trace resistance and low parasitic inductance.
 2. Use a common ground node for power ground and a different one for control ground to minimize the effects of ground noise. Connect these ground nodes at any place close to one of the ground pins of the IC.
 3. Use separate traces for the supply voltage of the power stage and the supply voltage of the analog stage.
 4. The sense trace connected to FB is signal trace. Keep these traces away from L1 and L2 nodes.

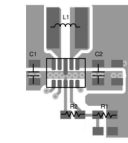
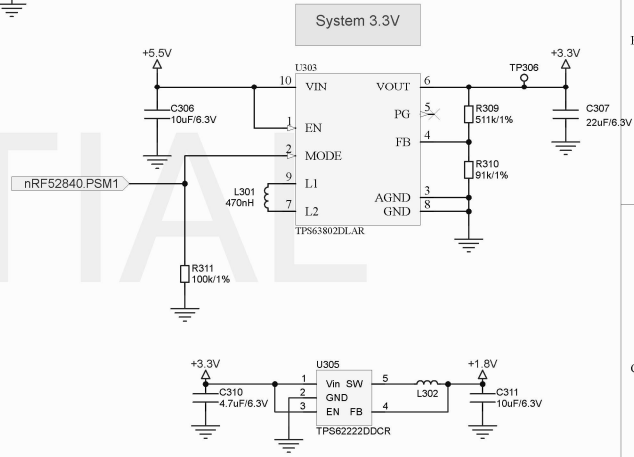


Figure 12-1. TPS63802 Layout



TPS73618 is an LDO that is pin compatible if the FB pin is no connected.

Layout should be setup so the FB trace can be cut for noise and efficiency testing

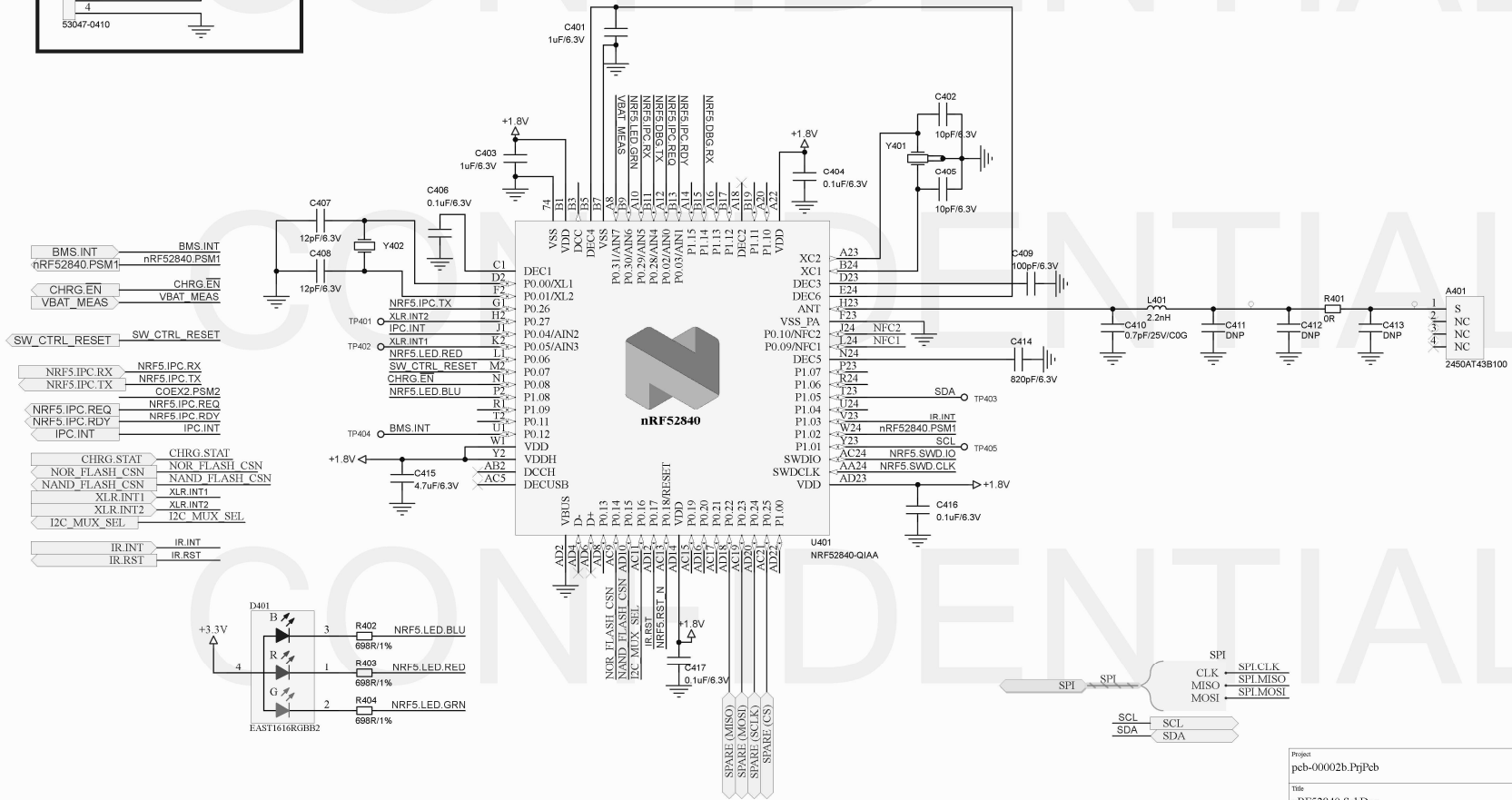
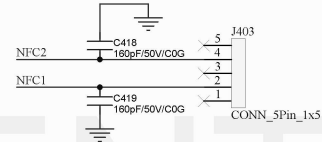
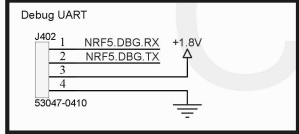
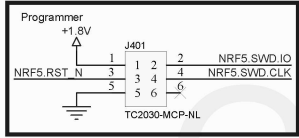
States:

- Normal - Battery voltage above 2.8V, power not plugged in
- Charging - Battery voltage below 4.2V, power plugged in
- Battery Full - Battery voltage at 4.2V, power plugged in
- UVLO Triggered - Battery voltage below 2.8V, power not plugged in
- UVLO Recovery - Battery voltage below 2.8V, power plugged in

R302 ~4.2uA of leakage from +BATT
R305+R307 ~3.2uA of Leakage from +BATT
R308 5uA of leakage from +5V_CHG
R309+R310 5.5uA of leakage from +3.3V

U301 16-30uA leakage in normal operation
U302 0.3uA leakage in normal operation
U303 11uA leakage in normal operation
U304 5.5uA leakage in normal operation
U305 15uA leakage in normal operation

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